INCH-POUND

MIL-PRF-32222(AT)
21 November 2006

PERFORMANCE SPECIFICATION

HEATER, VEHICULAR COMPARTMENT
MULTIFUEL BURNING 24 VOLT, DC
15,000 - 30,000 BTU/HR

This specification is approved for use by the U.S. Army Tank-automotive and Armaments Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification describes the performance requirements of the design, testing, manufacturing, and acceptance requirements for a forced air inlet personnel heater assembly.

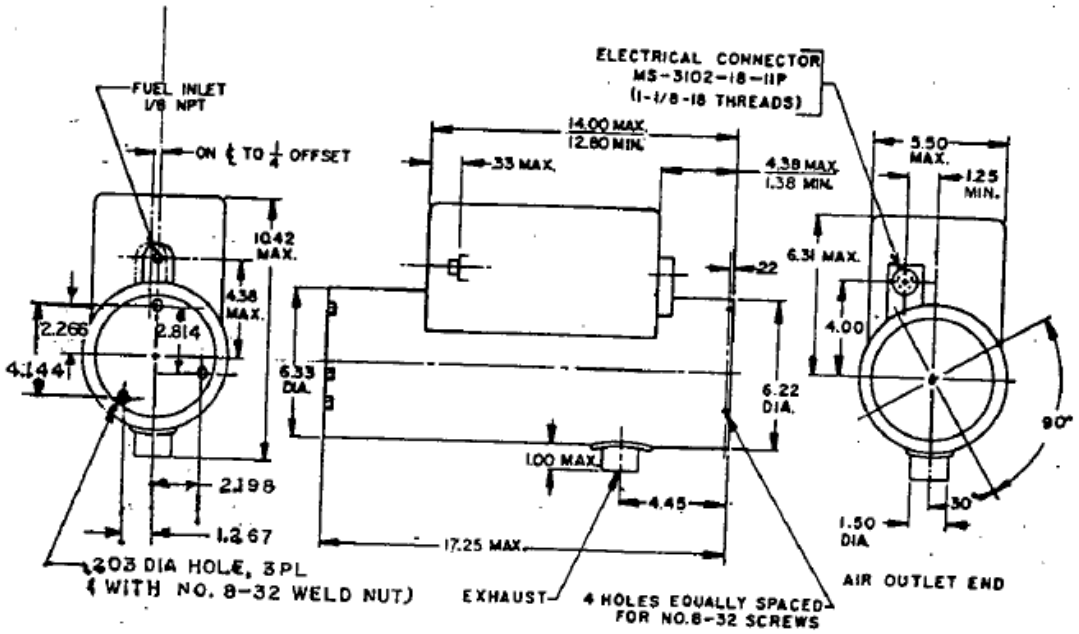
1.2 Item definition. The personnel heater, referred to as “heater”, should be a diesel fuel burning unit with an electrical control system and an electric blower.

1.2.1 Dimensions. The heater should conform to the dimensions shown on Figure 1. Arrangement of exhaust, inlet and outlet connections should be as shown on Figure 1.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Tank-automotive and Armaments Command, ATTN: AMSRD-TAR-E/268, MS-268, Warren, MI 48397-5000 by letter or emailed to standardization@tacom.army.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at http://assist.daps.dla.mil.

AMSC N/A FSC 2540

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NOTE: All dimensions in inches; unless otherwise specified tolerances are ± .030 inches, ± 1/2° on angles.

FIGURE 1. Heater Dimensions

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standard, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation form a part of this specification to the extent specified herein.

INTERNATIONAL STANDARDIZATION AGREEMENTS

ISO-10012	-	Equipment, Quality Assurance Requirements for Measuring Part I

(Unless otherwise indicated, copies of the above specification are available at http://www.iso.org/iso/en/ISOOnline.frontpage).

FEDERAL SPECIFICATIONS

GG-P-455	-	Plates and Foils, Photographic (Photo-sensitive Anodized Aluminum).

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FEDERAL STANDARDS

FED-STD-595	-	Colors Used in Government Procurement, Color Chip number 24084

COMMERCIAL ITEM DESCRIPTIONS

A-A-52557	-	Fuel Oil, Diesel
A-A-50271	-	Plates, Identification

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-5541	-	Chemical Conversion Coating on Aluminum and Aluminum Alloys
MIL-P-14105	-	Paint, Heat-resisting, (for Steel Surfaces)
MIL-DTL-19834	-	Plate, Identification, Metal Foil, Adhesive Backed.
MIL-P-53030	-	Primer Coating, Epoxy, Water Reducible, Lead and Chromate Free
MIL-PRF-62550E	-	Heater Assembly, Combustion Vehicular Compartment, 8.8/17.6 KW (30000/60000 BTU/HR), General Specification For

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-130	-	Identification Marking of US Military Property
MIL-HDBK-454	-	General Guidelines for Electronic Equipment
MIL-STD-461	-	Requirements for the Control of Electromagnetic Interference Characteristics of Subsystems and Equipment
MIL-STD-810	-	Environmental Test Methods
MIL-STD-889	-	Dissimilar Metals

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094, or at http://assist.daps.dla.mil/online/start/).

2.2 Non-Government publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DOD adapted should be the issue listed in the current DODISS and the supplement thereto, if applicable.

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AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME PTC 19.5	-	Performance Test Code

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 E. 47th Street, New York, NY 10017, or at www.asme.org).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D-2156-94	-	Smoke Density in Flue Gases from Burning Distillate Fuels, Test for
ASTM A967	-	Passivation Treatments for Stainless Steel Parts

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103, or at www.astm.com).

AIR MOVEMENT AND CONTROL ASSOCIATION (AMCA)

AMCA 210	-	Laboratory Methods of Testing Fans for Rating

(Application for copies should be addressed to the Air Movement and Control Association, 320 W. University Drive, Arlington Heights, IL 60004, or at www.amca.org).

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification should take precedence.

3. REQUIREMENTS

3.1 First article. Unless otherwise specified, the contractor shall furnish heaters which shall be subjected to first article inspection (see 4.4 and 6.2). First article inspection samples, properly marked with identifying information shall be representative of the unit to be furnished to the Government. All subsequent heaters delivered to the Government shall conform to these samples in all of their pertinent physical and performance attributes.

3.1.1 Materials. Materials shall be as specified herein, in applicable specifications, or drawings. Materials not specifically designated shall be suitable for use in heaters operating over specified ranges, without any change in properties that would result in operation of the units falling outside of the specified limits (see 6.4). The heater shall be fabricated from suitable corrosion-resistant materials or treated to prevent corrosion.

3.2 Dissimilar metals. Except where necessary to complete an electrical circuit, the compatibility of dissimilar metals shall be in accordance with MIL-STD-889.

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3.3 Design and construction. Heaters shall fit into current vehicle applications without modification to heater or vehicular components. Heaters shall be completely interchangeable as a unit with heaters of other manufacture used in the same application.

3.3.1 Weight. Weight of the heater shall not exceed 21 pounds.

3.3.2 Interchangeability. The assembly shall be so designed and fabricated that all component assemblies and parts are functionally and dimensionally interchangeable with like component assemblies and parts previously furnished by the same manufacturer. Construction shall be such that any part, except those furnished in matched sets or for which a selective fit is specified, may be installed, replaced, and adjusted without requiring modification.

3.4 Performance. The heater shall meet the performance requirements specified herein when mounted in any position from horizontal to vertical with the air inlet level with , or higher than, the air outlet (see 4.6).

3.4.1 Voltage. The heater shall start and meet performance requirements specified herein, in low and high heat, with an applied voltage of 19 to 30 volts direct current (V dc), 24 V dc nominal (see 4.6.1).

3.4.2 Current. At 24 V dc, maximum current drawn from the power source shall be 12 amperes (amps) starting and 8 amps running when the pre heaters are not activated. The current draw is increased by 5 amps maximum when the pre heaters are activated (approx. 45°F) (see 4.6.2.).

3.4.3 Fuel. The heater shall perform as specified herein using fuel conforming to CID A-A-52557 as follows (see 4.6.3):

Above plus 20°F	type DF-2
Plus 20° to minus 25°F	type DF-1
Below minus 25°F	type DF-A or JP-8

3.4.4 Fuel consumption. With any fuel pressure between 3 and 15 pounds per square inch gage (psig) at the fuel valve inlet, fuel consumption shall be as follows (see 4.6.4.):

High heat	0.049 pounds per minute (lbs/mil) maximum
Low heat	0.031 lbs/min maximum

3.4.5 Heat output. With a ventilating back pressure of 0.75 inch of water and an exhaust back pressure of 0.75 inch of water, heater output shall be as follows (see 4.6.5):

High heat	19.0 to 30.0 V dc, 27,000 British Thermal Units per hour (BTU/HR) minimum
Low heat	19.0 to 30.0 V dc, 15,000 BTU/HR ± 20 percent

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The air flow restrictions specified are to be set at 24 V dc input only and maintained in this position at all voltages.

3.4.6 Air discharge rate. At low heat and high heat, with 24 V dc applied the ventilating air discharge rate, based on standard air density of 0.075 pounds per cubic foot at 29.92 inches of mercury, 70ºF, shall be as follows (see 4.6.6):

Pressure	Discharge rate
At 0.00 inch (in.) water Ventilating back pressure Nominal and 0.00 in. water Exhaust back pressure nominal.	115 standard cubic feet per minute (scfm), minimum 130 scfm, nominal
At 0.75 in. water ventilating Back pressure nominal, and 0.75 in. water exhaust back pressure nominal.	80 scfm minimum 94

3.4.7 Overheat switch. The heater shall have an overheat switch which will automatically open a circuit and shut all fuel flow when the temperature of the discharge air reaches 400º plus or minus 75ºF (see 4.6.7).

3.4.8 Smoke density. After the heater has been operating for five minutes, minimum, exhaust gases shall not exceed a smoke spot number of three, except that at an ambient temperature of 73º ± 18º, and a voltage of 19 ± 0.5 V dc, the smoke spot number shall not exceed five. Smoke spot numbers are defined in (see 4.6.8).ASTM D-2156-94

3.4.9 Starting time. The starting time, measured from the time the heater is turned on until the igniter circuit is de-energized, shall not be greater that 4.0 minutes (see 4.6.9).

3.4.10 Purging time. The purging time, measured from the time the heater switch is turned off until the blowers shut off. Shall not be greater than 3.5 minutes (see 4.6.10).

3.4.11 Fuel leakage. There shall be no visible leakage from the heater fuel components during a period of five minutes, minimum, with the heater operating and 15 psig fuel pressure applied to the inlet (see 4.6.11).

3.4.12 Toxic fumes. Combustion and ventilating air in the heater shall be isolated from each other, after combustion. In operation, the heater shall not contaminate the ventilation air with products of combustion (see 4.6.12).

3.4.13 Endurance. The heater shall meet the requirements of 3.4.1 through 3.4.12 after 800 hours of operation; including 2400, minimum, start cycles. The following maintenance is permissible (see 4.6.13):

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a. Cleaning of fuel filters and screens.
b. Cleaning and adjustment of the flame detector switch, but not more that two times.
c. Replacement of the igniter, but not more than two times.
d. Replacement of the burner assembly if required, but no more frequently than once per sample during the test procedure.

3.4.14 Environmental conditions. The heater shall meet the requirements of 3.4.1 through 3.4.12 after being subjected to each environmental condition specified in 3.4.14.1 through 3.4.14.6 (see 4.6.14).

3.4.14.1 Operating temperature. The heater shall start and operate as specified herein at any temperature from plus 70ºF ± 5ºF to minus 55º F ± 5ºF, provided that fuel is supplied in liquid form (see 4.6.14.1 and 4.6.14.1.1.).

3.4.14.2 Storage temperature. The heater shall not be damaged by storage conditions ranging from minus 65º F to plus 160ºF (see 4.6.14.2).

3.4.14.3 Humidity. The heater shall withstand relative humidity up to 100 percent (see 4.6.14.3).

3.4.14.4 Corrosion. The heater shall function as specified herein after exposure to salt fog for 96 hours (see 4.6.14.4).

3.4.14.5 Shock. The heater shall function as specified herein after being subjected to 40g shock loading. Input to be measured at the interface of the heater and support bracket (see 4.6.14.5).

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3.4.14.6 Vibration. The heater shall start and operate as specified herein during and after sinusoidal vibration in accordance with Figure 2, for a period of 80 minutes minimum (including up to four resonance dwells of 13-1/3 minutes each) in each of three mutually perpendicular axes. In put to be measured at the interface of the heater and support bracket (see 4.6.14.6).

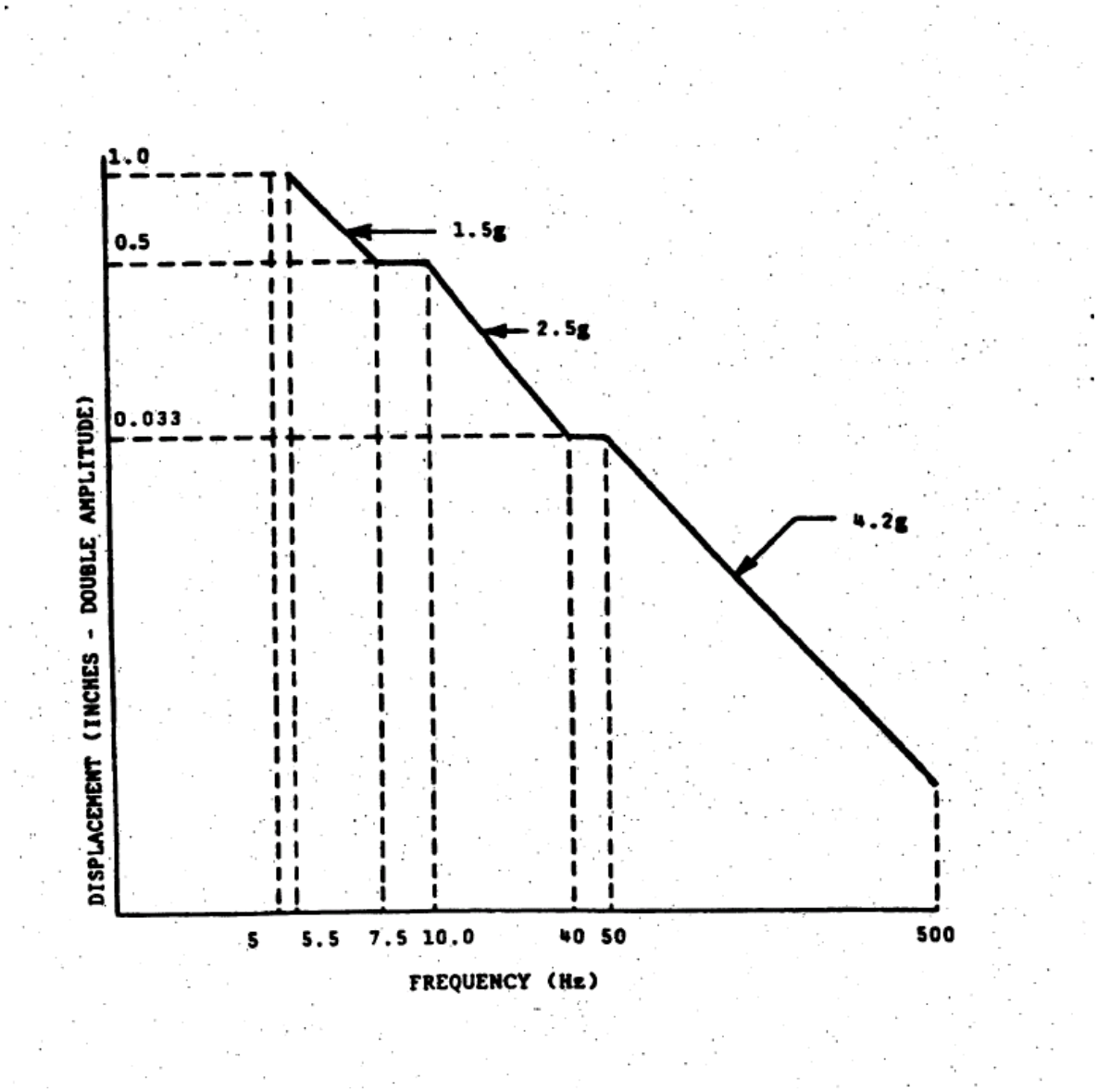


FIGURE 2. Vibration Curve

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3.4.14.7 Electromagnetic radiation. During any mode of operation, or while changing modes of operation, the heater shall meet the requirements of MIL-STD-461, as modified herein by Figures 3 thru 6 in appendix A. Emissions having a duration not exceeding one second, and recurring not more that once in three minutes, are exempt. Applicability of MIL-STD-461 and Figures 3 through 6 shall be as follows (see 4.6.14.7).

Requirement MIL-STD-461	Limits MIL-STD-461	Limits (MIL-PRF-32222)
CE01		Figure 3
CE04 (+28 V dc line)		Figure 4
RE02		Figure 5 and 6

3.5 Protective coating. All steel parts, except stainless steel, that are subjected to temperatures exceeding 200ºF, shall be painted with two coats of high temperature olive drab enamel conforming to MIL-P-14105. Surfaces shall be free of oil, grease, rust and mill scale before applying finish. All other painted steel parts shall be furnished in accordance with MIL-P-14105, or primed with primer conforming to MIL-P-53030. Heaters shall be painted flat or low luster green 383 or 24084 of FED-STD-595 or equivalent, and applied using best commercial practices.

3.5.1 Steel components. Unpainted steel components, other than stainless steel, shall have a protective coating as referenced in 3.5 above. Radiation surfaces may be aluminized steel.

3.5.2 Aluminum components. Aluminum components shall be anodized in accordance with MIL-C-5541, class 1A.

3.6 Marking and Identification plates.

3.6.1 Marking. Unless otherwise specified (see 6.2), heater shall be marked in accordance with MIL-STD-130.

3.6.2. Identification plates. As specified (see 6.2), nameplates shall be furnished in one of the following configurations:

a. Material and construction shall conform to (CID) A-A-50271, type III, composition A, class 2, or composition C. Composition C material shall conform to GG-P-455, type II, grade A class 1.

b. Adhesive backed aluminum foil shall conform to MIL-P-19834, type I, glossy: except backing sheet, which may be on paper, shall be removable intact by stripping without use of water or other solvents. Plate shall have an average adhesion, over bare and painted metal surfaces, of not less than 50 ounces per inch of width at 77º ± 5ºF.

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3.7 Workmanship. Heater shall be free of defects such as burrs, scratches, chips sharp edges, corrosion, scale and dirt. Workmanship shall also be in accordance with requirement (guideline) 9 of MIL-HDBK-454A.

3.8 Safety. Hot spots that could create a hazard to personnel shall be minimized and shall be identified on the drawings by the contractor. The heater shall be so designed that, with proper installation, heater exhaust gases cannot enter the vehicle personnel compartment.

4. VERIFICATION

4.1 Inspection equipment. Unless otherwise specified in the contract, the supplier is responsible for the provision and maintenance of all inspection equipment necessary to assure that supplies and services conform to contract requirements. Commercial, modified commercial, or supplier designed inspection equipment or measuring set-ups shall be capable of repetitive measurements to accuracy of 10 percent of the component tolerance. Calibration of inspection equipment shall be in accordance with ISO-10012-1.

4.2 Materials and construction. To determine conformance to 3.2 through 3.3.2, inspection and material certification records shall be maintained by the contractor. Records shall be subject to review by the Government and shall include date, part, or characteristic identification, inspection results, and disposition of lot (accepted or rejected). Corrective action taken on noted defects shall be subject to approval by the Government.

4.3 Classification of inspection:

a. First article inspection (see 4.5).
b. Quality conformance inspections (see 4.6).
    1. Examination (see 4.6.1).
    2. Acceptance tests (see 4.6.2.1).
    3. Control tests (see 4.6.2.2).

4.4 Inspection conditions. Unless otherwise specified, all inspections shall be conducted under the following conditions:

a. Air temperature	73º ± 18ºF
b. Barometric pressure	28.5 + 2.0 inches of mercury - 4.5
c. Relative humidity	50 ± 40 percent
d. Fuel	Diesel grade DF-2 (see 3.4.3)
e. Fuel pressure	8 ± 1 psi
f. Mounting position	As specified (see table II)

4.5 First article inspection. First article inspections are categorized as preproduction inspection and initial production inspection. Approval of the first article sample by the Government shall not relieve the contractor of the obligation to supply heaters that are fully representative of those inspected as a first article sample. Any changes or deviation of the production units from the first article sample shall be subject to the approval of the contracting officer.

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4.5.1 Preproduction inspection. When specified (see 6.2), the preproduction sample shall consist of three heaters. Preproduction inspection samples shall be examined per table I and inspected per table II under the surveillance of the Government inspector.

4.5.2 Initial production inspection. During fabrication of the first production heater, an in-process inspection shall be conducted by the contractor under the surveillance of the Government inspector. Inspection shall be made at the contractor's facility prior to application of primer and paint. Processing and welding procedures, quality system and inspection records shall be reviewed and evaluated during this inspection.

4.5.2.1 Test routine. Following in-process inspection, three heaters that have been produced with production materials, equipment and procedures shall be subjected to inspection by the contractor under the surveillance of the Government inspector. Initial production units shall be examined per table I and inspected per table II.

4.5.3 First article inspection failure. Failure of any first article sample to pass specified examinations or tests shall be cause for refusal to grant first article approval until corrective action by the contractor has been approved by the Government.

4.5.4 Heater disposition. After completion of inspection, the first production heater shall remain at the manufacturing facility, as a production sample, and be the last heater shipped on the contract. The heater may be released sooner at the discretion of the contracting officer. The contractor shall service and maintain the heater during this period.

4.6 Lot formation. An inspection lot shall consist of all the heaters of one type and part number, from an identifiable production period, from one manufacturer, submitted at on time for acceptance.

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4.6.1 Examination. Units under control test shall be examined for defects specified by Table I.

TABLE I. Classification of defects.

Category	Defect	Method of Examination
Major		
101	Improper materials (see 3.2)	Visual
102	Dimensions affecting interchangeability not within tolerance	SIE 1/
Minor		
201	Weight over maximum allowed (see 3.3.1)	Scale
202	Improper protective coating (see 3.5)	Visual
203	Improper marking of identification (see 3.6)	Visual

1/ SIE = Standard Inspection Equipment.

4.6.2 Quality conformance tests.

4.6.2.1 Acceptance test. Each heater shall be subjected to the tests specified in table II.

4.6.2.2 Control tests. Control tests shall be conducted on one heater of each lot of 600 units consecutively produced, except that not more that one unit shall be tested in a three month period, not less that one unit in a six month period. The heaters shall be examined per table I and tested per table II.

4.6.2.3 Failure. Failure of any heater to pass any of the specified inspections shall be cause for the Government to refuse acceptance of the production quantity represented, until action taken by the contractor to correct defects and prevent recurrence has been approved by the Government.

4.7 Conformance verification. Except as otherwise specified, performance tests shall be conducted with heater mounted in position specified in table II, at both high and low heat, at input voltages of 19.0 ± 0.5 V dc, 24.0 ± 0.5 V dc and 30.0 ± 0.5 V dc. Acceptance inspection shall be conducted at 24.0 ± 0.5 V dc only. Prior to each start, heater shall be cooled for a minimum of 10 minutes.

4.7.1 Voltage test. Applied voltage shall be monitored throughout performance testing to determine conformance to 3.4.1.

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4.7.2. Current test. Current draw from the power source shall be monitored during performance testing, while starting and running at ambient temperatures above and below 45ºF, to determine conformance to 3.4.2. Running current shall be measured after the igniter is de-energized.

4.7.3 Fuel test. Acceptable performance during testing as specified in 4.7.13 and 4.7.13.1, using the fuel types specified, shall determine conformance to 3.4.3.

4.7.4 Fuel consumption test. Fuel flow shall be measured with a fuel pressure of 15 (+0, -1) psig applied to the fuel valve inlet and subsequently with a pressure of 3 (+1, -0) psig, similarly applied, to determine conformance to 3.4.4.

4.7.5 Heat output test. With a ventilating back pressure of 0.75 inch of water and an exhaust back pressure of 0.75 inch of water (both established at 24 V dc input only), thermocouples measuring input and output air temperatures; and an air flow meter measuring output air flow, the heater shall be operated until the outlet temperature has stabilized. Output thermocouples shall be located 42 ± 1 inch from the heater, within the duct used in 4.7.6. When testing is performed at ambient temperatures above 60ºF, and the overheat switch cycles, the switch may be bypassed. Bypassing is only allowed when outlet temperature exceeds 325ºF and not more that 475ºF. Heat output shall be calculated, using the following equation, to determine conformance to 3.4.5.

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TABLE II. Classification of inspection.

First article and first production							
			Samples			Acceptance	Control
Title	Requirement	Test	1	2	3	test (100%)	test
			Horiz	Vert	Horiz	Horiz	Horiz
Performance	3.4	4.7	x	x	x	x	
Voltage	3.4.1	4.7.1	x	x	x	x	
Current	3.4.2	4.7.2	x	x	x	x	
Fuel	3.4.3	4.7.3	x	x	x	x	
Fuel consumption	3.4.4	4.7.4	x	x	x	x	
Heat output	3.4.5	4.7.5	x	x	x	x	
Air discharge rate	3.4.6	4.7.6	x	x	x	x	
Overheat switch	3.4.7	4.7.7	x	x	x	x	
Smoke density	3.4.8	4.7.8	x	x	x	x	
Starting time	3.4.9	4.7.9	x	x	x	x	
Purging time	3.4.10	4.7.10	x	x	x	x	
Fuel leakage	3.4.11	4.7.11	x	x	x	x	
Toxic fumes	3.4.12	4.7.12	x	x	x	x	
Endurance	3.4.13	4.7.13	x	x			
Endurance control		4.7.13.1					x
Low temperature operation	3.4.14.1	4.7.14.1	x	x	x		x
High temperature operation		4.7.14.1.1	x	x	x		
Storage temperature	3.4.14.2	4.7.14.2	x		x		
Humidity	3.4.14.3	4.7.14.3		x			
Corrosion	3.4.14.4	4.7.14.4	x				
Shock	3.4.14.5	4.7.14.5	x	x			x
Vibration	3.4.14.6	4.7.14.6		x	x		x
Electromagnetic radiation	3.4.14.7	4.7.14.7			x		

H = V x d x CP x T
H = Heat output in BTU/hour
V = Air flow in cu. Fr. Per hour (Standard air)
d = Density of air at 70ºF, 0.0750 lbs. per cu. Ft. (standard air)
Cp = Specific heat of air (average) as corrected for temperature and moisture content.
T = (outlet temperature, in ºF) minus (inlet temperature, in ºF)

4.7.6 Air discharge rate test. To determine conformance to 3.4.6, measure the ventilating air discharge velocity with 24 V dc input and back pressure as specified in 3.4.6 utilizing AMCA Standard 210-74, (pitot traverse in outlet duct, or a venture complying with ASME Power Test Code 19.5.4 - 1959) back pressure may be controlled by using either variable or fixed restrictions sized on a nominal heater under nominal ambient conditions run at 24 V dc input in the high heat range. The air discharge rate shall be calculated using the methods defined in AMCA Standard for 24 V dc. Unrestricted air flow measurements are not required during 100% acceptance testing.

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4.7.7 Overheat Switch test. To determine conformance to 3.4.7, operate the heater at high heat into the duct used in 4.7.6, restricting the air flow to cause a temperature buildup in the duct. Measure the temperature of the ventilating air at a point 42 ± 1 inch from the heater, within the duct. The overheat switch shall break the circuit and cut off the fuel when the air temperature in the duct exceeds 325ºF but is less that 475ºF. This test is to be conducted at room ambient temperature only (73º ± 18ºF).

Note: Heater radiation effects should be negated by appropriate shielding of heat sensing devices.

4.7.8 Smoke density test. To determine conformance to 3.4.8, using the test methods set forth in ASTM-D-2156-94, measure the smoke density of the heater exhaust gases within 6 ± 1 minute after ignition. The smoke density shall not exceed a smoke spot number of three (3) above 24 V dc, and a smoke spot number of five (5) at 19 ± 0.5 V dc.

4.7.9 Starting time test. To determine conformance to 3.4.9, measure and record the time from the starting switch being turned on until the igniter is de-energized. The elapsed time shall not exceed 4.0 minutes. This test shall be conducted at low heat only.

4.7.10 Purging time test. To determine conformance to 3.4.10, after test of 4.7.9, turn the temperature control to high heat. Turn off the starting switch. Measure the time from the starting switch being turned off until the blowers shut off. The elapsed time shall not exceed 3.5 minutes.

4.7.11 Fuel leakage test. The heater shall be operated for a minimum of 5 minutes at a fuel pressure of 14.0 ± 1 psi. The heater shall subsequently be examined for evidence of fuel leakage to determine conformance to 3.4.11.

4.7.12 Toxic fumes test. To determine conformance to 3.4.12, for heaters with a separate combustion air inlet, the inlet shall be sealed and 3 inches of water pressure applied to the exhaust outlet. Leakage shall not exceed one scfm. Prior to assembly of the heat exchanger into the heater, it shall be tested for leakage at 10 psig under water. No leakage shall be permitted. After assembly, 10 psig leakage tests need not be repeated during sub sequent testing, except each first article test sample and control test sample shall be retested at the completion of all other testing. No leakage shall be permitted.

4.7.13 Endurance test. To determine conformance to 3.4.13, the heater shall be operated for a total of 800 hours "ON" time (ignite and run) (see 4.6.5.2.1). Operation shall be in 30 minute cycles consisting of 20 minutes "ON" time and 10 minutes "OFF" time (purge and cool).

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The total 800 hours of operation shall consist of only "ON" time will not be included in compilation of operating time. Operation shall consist of the following phases:

Phase I - Using DF-2 fuel at room ambient (73º ± 18ºF),

a. Set controls at low heat and operate heater in cycles previously described for 8 hours with an input voltage of 19.0 ± 0.5 V dc.
b. Repeat step a using voltage input of 24.0 ± 0.5 V dc.
c. Repeat step a using voltage input of 30.0 ± 0.5 V dc.
d. Reset temperature control to high heat and repeat steps a, b and c.
e. Repeat steps a through d until 600 thirty minute cycles are completed (200 "ON" hours).
f. Test heater as specified in 4.7.1 through 4.7.12.

Phase II - Using DF-1 fuel at minus 25º ± 5ºF (both fuel and heater)
Repeat steps a through f, except 300 thirty minute cycles (100 "ON" hours) shall be completed.

Phase III - Using DF-A or JP8 fuel at minus 55º ± 5ºF (both fuel and heater) repeat steps a through f, except 300 thirty minute cycles (100 "ON" hours) shall be completed.

Phase IV - Repeat phases I through III for an additional 400 hours.

4.7.13.1 Control endurance test. For control test purposes, the "ON" time endurance testing of 4.7.13 shall be reduced to 150 hours. Test shall consist of 50 hours each of phases I, II and III. Determine conformance to 3.4.13.

4.7.14 Environmental conditions tests.

4.7.14.1 Low temperature operation test. To determine conformance to 3.4.14.1, the heater shall be subjected to the low temperature test specified in MIL-STD-810, method 502.4, procedure II. Temperature shall be maintained at minus 65ºF for a minimum of 24 hours. Subsequently, temperature shall be raised to minus 55ºF and heater tested as specified in 4.7.1 through 4.7.12 using DF-A of JP8 fuel. During testing, stabilized combustion shall be maintained for 30 minutes on low heat and again for 30 minutes on high heat. Prior to any test, heater shall be stabilized at minus 55ºF before each ignition. Subsequently, testing shall be repeated at minus 25ºF using DF-1 fuel, and at plus 20°F using DF-2 fuel. At each of the three temperatures, the tests specified in 4.7.9 and 4.7.10 shall be repeated twice more, for a total of three times.

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4.7.14.1.1 High temperature operation test. To determine conformance to 3.4.14.1, the heater shall be subjected to the high temperature test specified in MIL-STD-810, method 501.4, procedure II, except that steps 4 and 5 of the procedure shall be omitted.

4.7.14.2 Storage temperature test. This test is conducted in conjunction with 4.7.14.1 and 4.7.14.1.1.

4.7.14.3 Humidity test. To determine conformance to 3.4.14.3, the heater shall be tested as specified in MIL-STD-810 method 507.4, (omitting step 4). After exposure, heater shall be tested as specified in 4.6.1 through 4.6.12.

4.7.14.4 Corrosion resistance test. To determine conformance to 3.4.14.4 the heater shall be tested as specified in MIL-STD-810 method 509.4. Subsequently, heater shall be tested as specified in 4.7.1 through 4.7.12.

4.7.14.5 Shock test. To determine conformance to 3.4.14.5, the heater, with no fuel applies, shall be subjected to the half sine shock pulse configuration test specified in MIL-STD-810, method 516.5, procedure I, amplitude 40g's, duration 18ms. During testing, heater shall be subjected to input power and monitored for intermittent conditions. Subsequently, heater shall be tested as specified in 4.7.1 through 4.7.12.

4.7.14.6 Vibration test. To determine conformance to 3.4.14.6, the heater shall be rigidly mounted and subjected to vibration as specified in MIL-STD-810, method 514.5, procedure I, using the curve shown on figure 1, at ambient temperature. During testing, heater shall be started, purged, operated at both heats, and monitored for intermittent operation. A resonant frequency search shall be conducted along each axis, with the heater operating at low heat, over a range of 5 to 500 Hz. The test schedule shall include 13 1/3 minutes dwell time at each resonance, to a maximum of four. If more that four resonances are present, the most severe shall be used. After determining resonance at low heat, the specified resonance dwell and sinusoidal cycling tests shall be conducted. The total cycling time per axis shall be 80 minutes including the resonance dwell time accumulated. Subsequently, heater shall be tested as specified in 4.7.1 through 4.7.12.

4.7.14.7 Electromagnetic radiation test. To determine conformance to 3.4.14.7, the heater shall be tested as specified in MIL-STD-462, figures 2 through 5.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

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6. NOTES

6.1 Intended use. The forced air inlet personnel heater furnished under this specification is intended to supply warm air to the crew compartment of military vehicles.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Applicable drawing (see 3.3).
c. Any special marking requirements (see 3.6.1).
d. Configuration of identification plates (see 3.6.2).
e. Applicable level of packaging, packing, and marking (see 5.1).
f. Preproduction inspection, if required (see 4.4.1).
g. Initial production inspection, if not required (see 4.4.2).

6.3 Recycled materials. The use of recycled materials which meet the requirements of the applicable material specifications without jeopardizing the intended use of the item should be encouraged (see 3.2).

6.4 Subject term (key word) listing.

Air flow
Combustion vapor
Diagnostics, self
Over pressure
Smoke density
Ventilation

6.5 Suppression data. This specification in lieu of MIL-PRF-62550D(AT), Type I Class I. MIL-PRF-62550/3(AT), 30 May 1997, which classified the heater specified herein as Type I, Class I.

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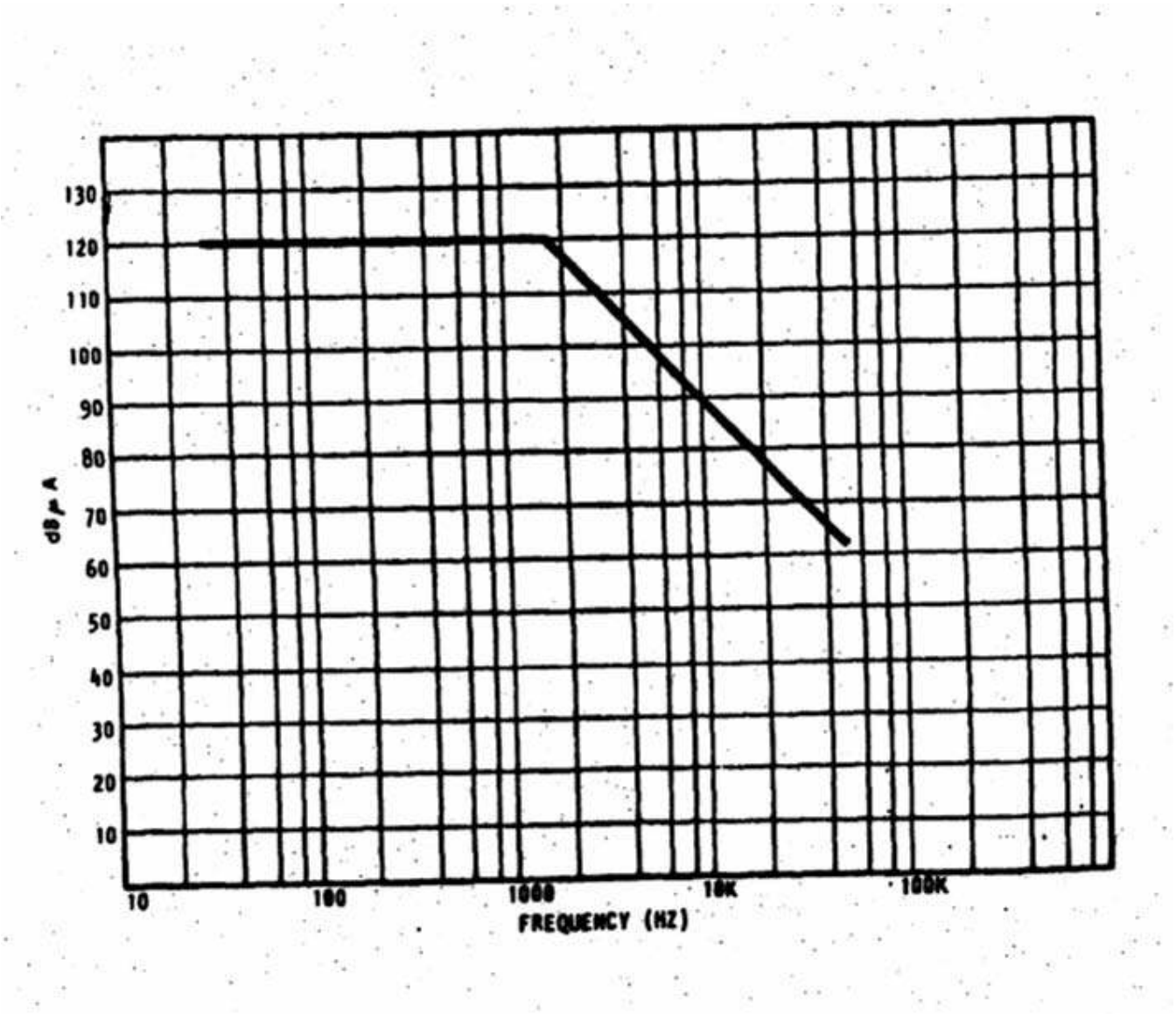


FIGURE A1. Conducted Emissions (CE01) + 28 VDC line

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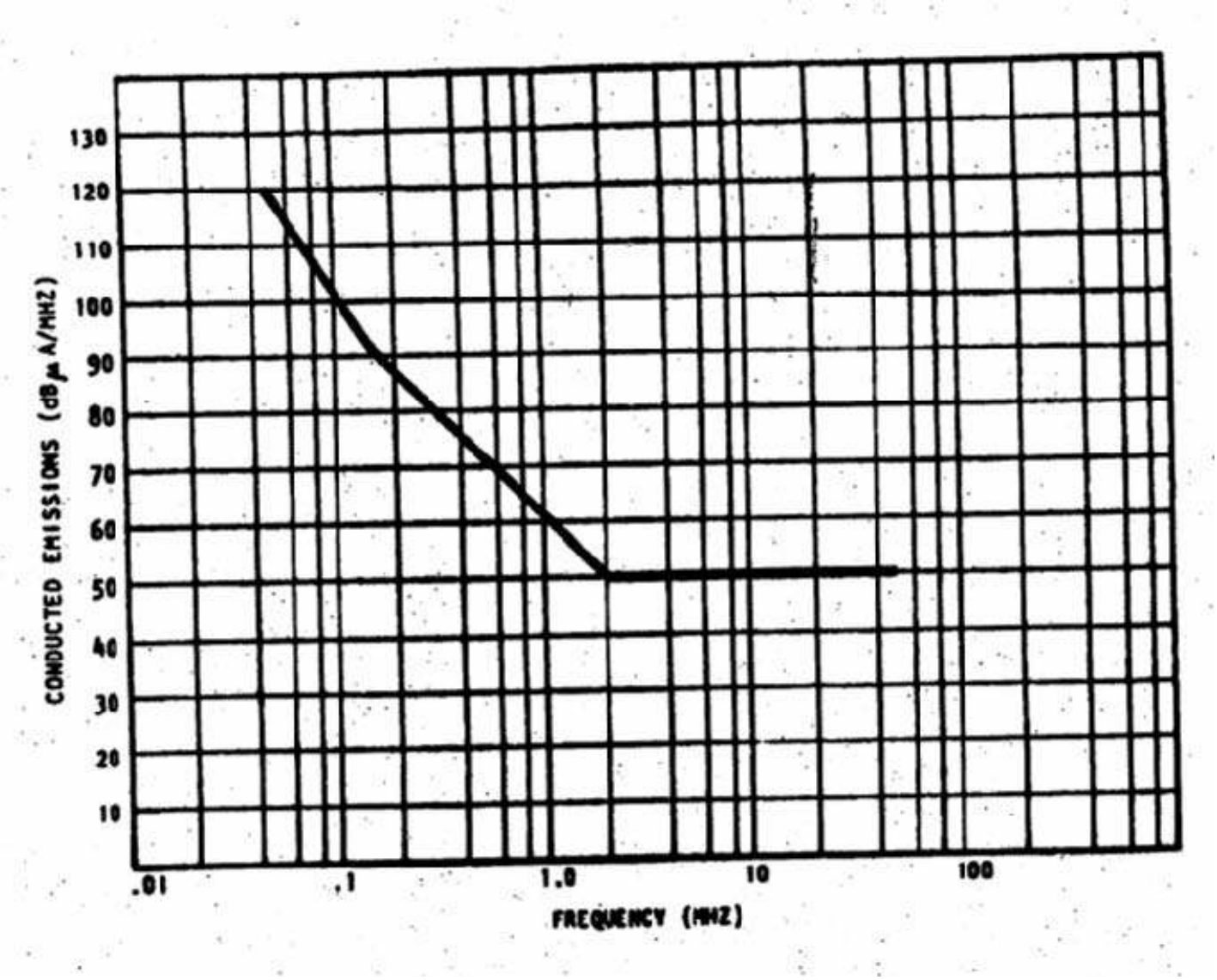


FIGURE A2. Conducted Emissions (CE04) + 28 VDC Lead

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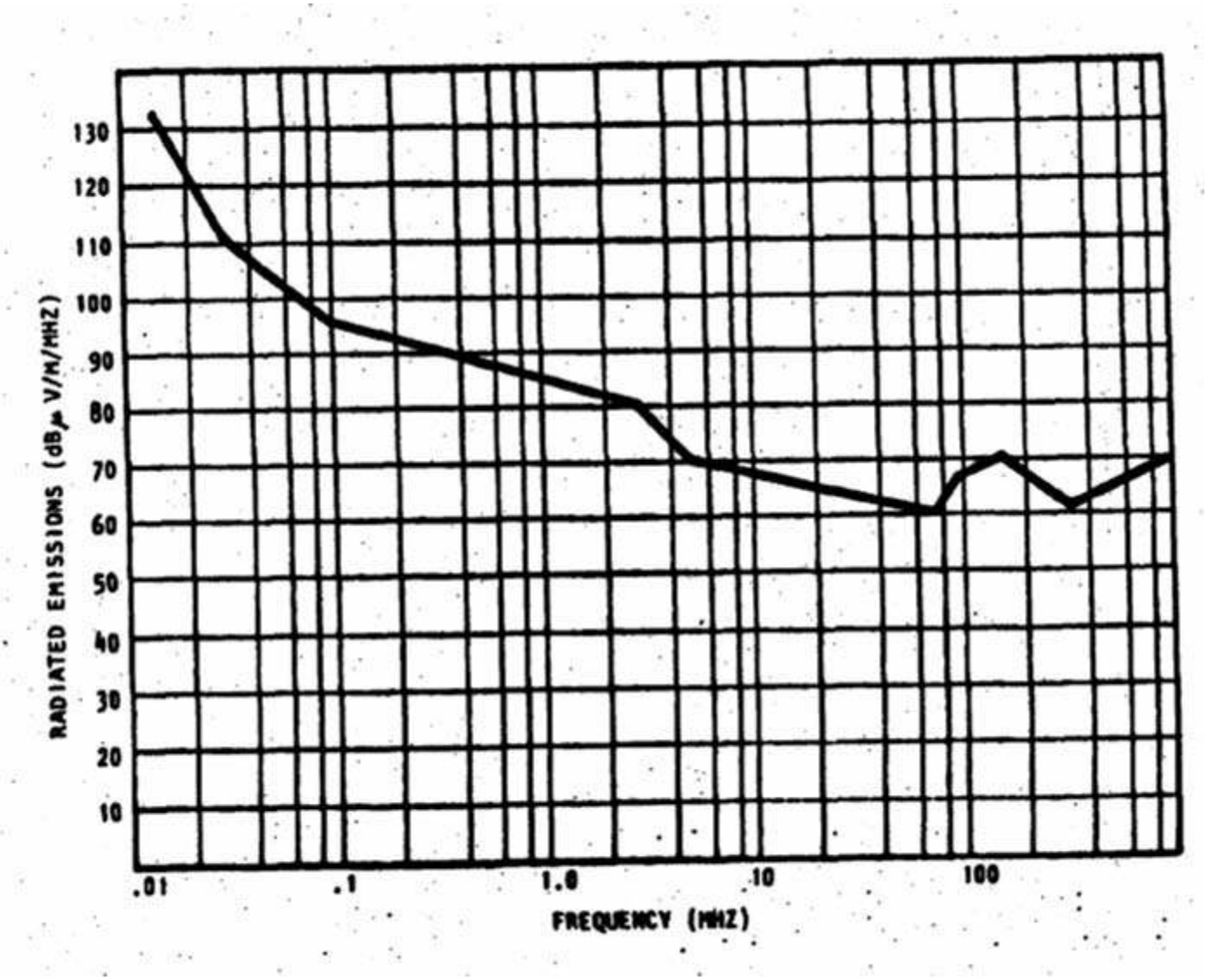


FIGURE A3. Broadband Radiated Emissions (RE02), Flame Detect Switch – Switching Transients

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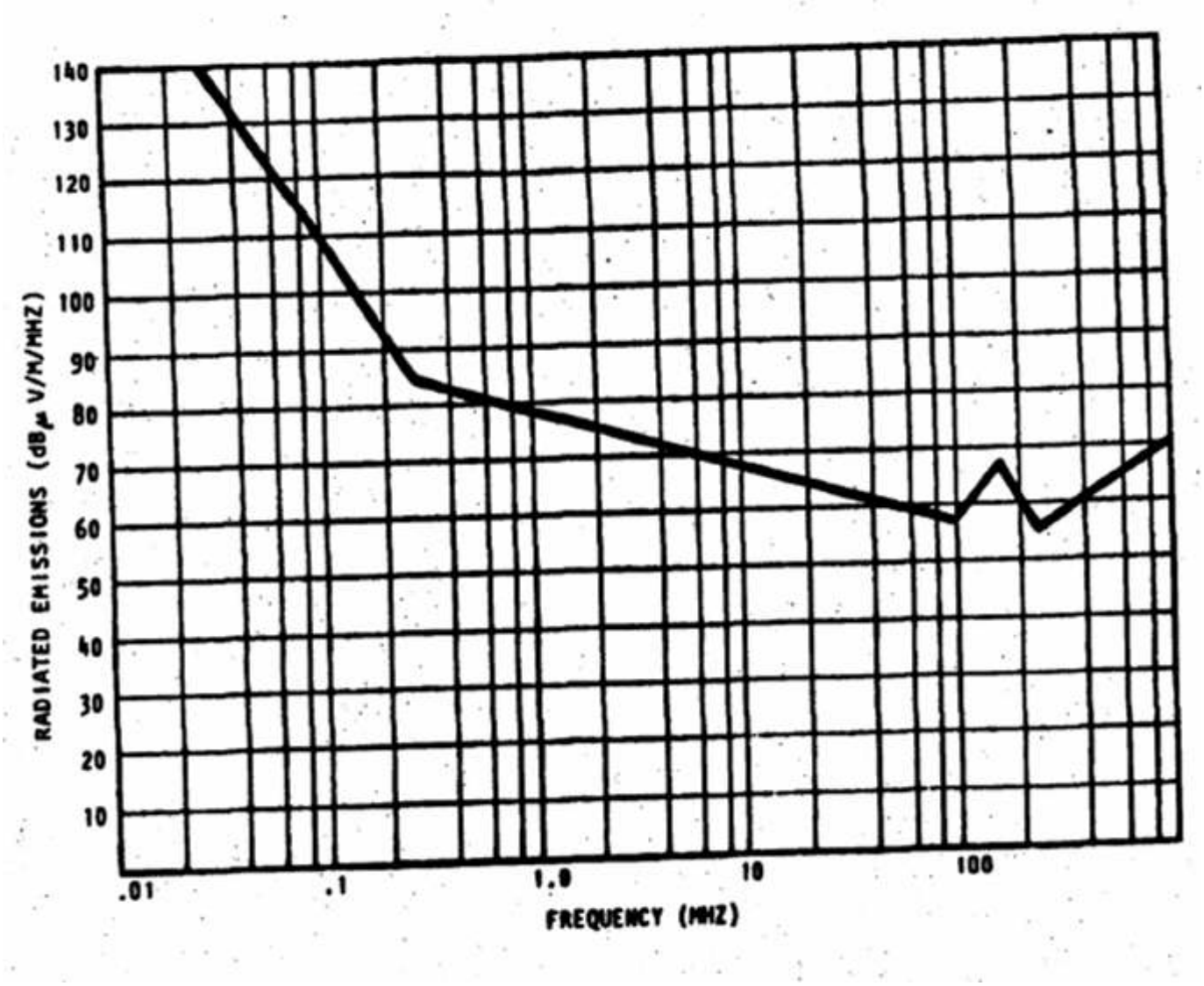


FIGURE A4. Broadband Radiated Emissions (RE02) Heater Operating – Steady State

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Preparing Activity:
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